

OS-PRACTICAL-1

1. Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that 1. Accept a Linux command as input. 2. Create a child process using fork(). 3. Execute the command in the child process using an appropriate exec() system call. 4. Allow the parent process to wait for the child using wait (). 5. Display the Process ID (PID) of both parent and child processes.

- Created a file named practical1.c using

```
rishita@Rishitha:~$ nano practical1.c|
```

- Compiled and ran the program where the command “ls” is entered.

```
rishita@Rishitha:~$ nano practical1.c
rishita@Rishitha:~$ gcc practical1.c
rishita@Rishitha:~$ ./a.out
ls
ls
```

- The Child PID has been returned followed by its command execution while the parent process waited till the child process completes.

```
rishita@Rishitha:~$ nano practical1.c
rishita@Rishitha:~$ gcc practical1.c
rishita@Rishitha:~$ ./a.out
ls
ls
Child PID is = 559
a.out  b1      exec2.c  exec5.c  files.c.save  fork.c      linux_task4
a2     c2      exec3.c  f1       files1.c      fork.c.save newfile
a3     exec1.c exec4.c  files.c  files1.c.save fork3.c     open2.c
Parent PID is = 323
```

Using Linux terminal commands (uname, lscpu, lsblk, ps, top), investigated the relationship between hardware resources and operating system services

```

rishita@Rishitha:~$ uname
Linux
rishita@Rishitha:~$ lscpu
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:          39 bits physical, 48 bits virtual
Byte Order:             Little Endian
CPU(s):                 12
On-line CPU(s) list:   0-11
Vendor ID:              GenuineIntel
Model name:             12th Gen Intel(R) Core(TM) i5-1235U
CPU family:            6
Model:                  154
Thread(s) per core:    2
Core(s) per socket:    6
Socket(s):              1
Stepping:               4
BogoMIPS:               4991.99
Flags:                  fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss ht syscall nx pdpe1gb rdtsc
                        cp lm constant_tsc rep_good nopl xtopology tsc_reliable nonstop_tsc cpuid tsc_known_freq pni pclmulqdq vmx ssse3 fma cx16 pcid
                        sse4_1 sse4_2 x2apic movbe popcnt tsc_deadline_timer aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch ssbd ibrs i
                        bpb stibp ibrs_enhanced tpr_shadow ept vpid ept_ad fsgsbase tsc_adjust bmi1 avx2 smep bmi2 erms invpcid rdtseed adx smap clflush
                        opt clwb sha_ni xsaveopt xsavec xgetbv1 xsaves avx_vnni vnni umip waitpkg gfni vaes vpclmulqdq rdpid movdiri movdir64b fsrm md_
                        clear serialize ibt flush_lld arch_capabilities

Virtualization features:
Virtualization:         VT-x
Hypervisor vendor:     Microsoft
Virtualization type:    full
Caches (sum of all):
  L1d:                   288 KiB (6 instances)
  L1i:                   192 KiB (6 instances)
  L2:                    7.5 MiB (6 instances)
  L3:                    12 MiB (1 instance)
NUMA:
NUMA node(s):           1
NUMA node0 CPU(s):     0-11
Vulnerabilities:
Gather data sampling:   Not affected
Ghostwrite:            Not affected
Indirect target selection: Not affected
Intel_mitihit:         Not affected
L1tf:                  Not affected
Mds:                   Not affected
Meltdown:              Not affected
Mmio stale data:       Not affected
Old microcode:         Not affected
Reg file data sampling: Mitigation; Clear Register File
Retbleed:              Mitigation; Enhanced IBRS
Spec_rstack_overflow:  Not affected
Spec_store_bypass:     Mitigation; Speculative Store Bypass disabled via prctl
Spectre_v1:            Mitigation; usercopy/swapgs barriers and __user pointer sanitization
Spectre_v2:            Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBRSE-eIBRS SW sequence; BHI BHI_DIS_S
Srbds:                 Not affected
Tsa:                   Not affected
Tsx_async_abort:       Not affected
Vmscape:               Not affected
rishita@Rishitha:~$ |

```

```

vmscape:
rishita@Rishitha:~$ lsblk
NAME MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda   8:0    0 356.9M  1 disk
sdb   8:16   0 159.4M  1 disk
sdc   8:32   0    2G    0 disk [SWAP]
sdd   8:48   0    1T    0 disk /mnt/wslg/distro
/
rishita@Rishitha:~$ |

```

```

rishita@Rishitha:~$ ps
  PID TTY          TIME CMD
  323 pts/0        00:00:00 bash
  696 pts/0        00:00:00 ps
rishita@Rishitha:~$ |

```

```

rishita@Rishitha:~$ top
top - 05:13:54 up 15 min, 1 user, load average: 0.00, 0.01, 0.00
Tasks: 24 total, 1 running, 23 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.1 us, 0.1 sy, 0.0 ni, 99.5 id, 0.2 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7784.6 total, 7203.1 free, 545.0 used, 187.5 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7239.6 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
    1 root        20   0   24116   15356   11588 S   0.3   0.2   0:01.16 systemd
   50 root        19  -1   50400   16716   15432 S   0.3   0.2   0:00.31 systemd-journal
    2 root        20   0    3180    2212    2072 S   0.0   0.0   0:00.01 init-systemd(Ub
    7 root        20   0    3180    2044    1940 S   0.0   0.0   0:00.00 init
   82 systemd+   20   0   22416   14448   11956 S   0.0   0.2   0:00.14 systemd-resolve
   89 root        20   0   35372   12352    9256 S   0.0   0.2   0:00.76 systemd-udev
  111 root        20   0    2888    1948    1820 S   0.0   0.0   0:00.12 chronyd-starter
  112 root        20   0    4472    3148    2896 S   0.0   0.0   0:00.01 cron
  113 message+   20   0    8792    5500    4828 S   0.0   0.1   0:00.10 dbus-daemon
  119 root        20   0   44096   29864   14724 S   0.0   0.4   0:00.58 networkd-dispat
  127 root        20   0   18440    9468    8228 S   0.0   0.1   0:00.21 systemd-logind
  184 syslog     20   0  220548    5756   4604 S   0.0   0.1   0:00.15 rsyslogd
  223 _chrony     20   0   23588   11068    7148 S   0.0   0.1   0:00.09 chronyd
  233 _chrony     20   0   12096    2452    1804 S   0.0   0.0   0:00.01 chronyd
  288 root        20   0    5492    2700    2516 S   0.0   0.0   0:00.01 agetty
  292 root        20   0  123460   32676   18072 S   0.0   0.4   0:00.34 unattended-upgr
  320 root        20   0    3184    1116    980 S   0.0   0.0   0:00.00 SessionLeader
  321 root        20   0    3200    1252    1108 S   0.0   0.0   0:00.10 Relay(323)
  323 rishita     20   0    6268    5532    3860 S   0.0   0.1   0:00.13 bash
  324 root        20   0    8648    5544    4700 S   0.0   0.1   0:00.01 login
  389 rishita     20   0   22348   13620   10988 S   0.0   0.2   0:00.16 systemd
  391 rishita     20   0   22924    4092    2088 S   0.0   0.1   0:00.00 (sd-pam)
  421 rishita     20   0    6136    5436    3864 S   0.0   0.1   0:00.06 bash
  706 rishita     20   0    8192    6064    3940 R   0.0   0.1   0:00.01 top

```

1. uname

It displays basic information about the Linux operating system and kernel.

2. lscpu

It displays CPU information such as the number of CPUs, cores, architecture, and processor details.

3. lsblk

It displays the storage devices and partitions available in the system.

4. ps

It displays the processes currently running in the system.

5. top

It display running processes along with their CPU and memory usage in real time.

Hardware and the Operating System :

The operating system acts as a link between applications and computer hardware.

CPU: The OS schedules processes and assigns them CPU time.

Memory: The OS allocates RAM to processes and keeps their memory spaces separate.

Storage: The OS manages storage devices and organizes data into files and directories.

I/O Devices: The OS uses device drivers to communicate with devices such as keyboards, displays, and network cards.